Max Ellis

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Summary

High-velocity Senior Software Engineer with a proven track record of architecting scalable data platforms and automating mission-critical infrastructure. Expert in backend development for distributed systems, utilizing Java, Snowflake, and AWS to process massive data volumes. Recognized for "smoke jumping" into critical projects to restore stability, performance, and reliability. Passionate about building efficient, fault-tolerant systems that empower users.

Work Experience

Senior Software Engineer **Act-On Software, Portland** **June 2025 – Present**

- **AI Product Replacement:** Sole backend engineer for ground-up replacement of a partner AI product under a 3-month deadline. Architected multi-agent, multi-source RAG infrastructure using AWS Bedrock, including web crawler with Cloudflare and authentication handling, and queuing system. Delivered initial parity on deadline, subsequently expanded to many-to-many agent/source mapping.
- **Email Performance Optimization:** Built a subject line optimizer that feeds top and bottom performing subject lines into a Bedrock agent to find patterns and propose alternatives, which drove a 17% increase in customer email open rates.
- **High-Scale Data Architecture:** Architected and implemented a scalable schema-agnostic Custom Objects feature capable of ingesting and processing over 1 million complex records per 10 minutes. Leveraged Snowpipe streaming for real-time ingestion and Kafka integrations to send changesets to downstream consumers, ensuring data integrity for customer analytics.
- **Operational Excellence:** Proposed OpenFlow as a solution to replicate Postgres data to Snowflake. Used Terraform to provision VPCs, subnets, security groups, IAM roles, and Hashicorp Vault rules to support OpenFlow. Automated the connector creation process, reducing manual steps and reducing setup time for new connectors from several hours to less than 20 minutes.
- **Mentorship:** Mentored junior and mid-level engineers on an engineering best practices and AI-augmented development workflows, including hands-on training with Claude and other LLM tooling.

Mid-level Software Engineer **Act-On Software, Portland** **March 2024 – June 2025**

- **Legacy Modernization:** Spearheaded the upgrade of a mission-critical dataflow service from Java 8 to Java 21. Consolidated distributed H2 databases into a centralized PostgreSQL cluster, improving data visibility and system reliability.
- **Pipeline Engineering:** Engineered multiple data processing pipelines using Spring Cloud Data Flow to seamlessly integrate Snowflake, S3, and internal microservices.

Junior Software Engineer **Act-On Software, Portland** **November 2022 – March 2024**

- **Performance Optimization:** Optimized data handling strategies for a data ingestion microservice, eliminating data duplication errors and drastically reducing response times from 8 minutes to 300 milliseconds.
- **Process Improvement:** Identified and fixed a critical UTF-8 corruption issue affecting 12% of customer data, preventing significant customer impact.

Education

Master of Science, Computing Science, University of Alberta, Edmonton **June 2022**
Bachelor of Science, Computer Science, Washington State University, Vancouver **May 2019**

Technical Skills

Languages:	Java, Spring, SQL, Python, Node.js, React
Cloud & Infra:	AWS (EC2, S3, Dynamo, API Gateway, Lambda, etc.), Terraform, Kubernetes, Docker
Data & DB:	Snowflake, OpenFlow, PostgreSQL, Kafka, JDBC, Spring Cloud Data Flow
Tools:	Bitbucket Pipelines, IntelliJ IDEA, HashiCorp Vault, Splunk, New Relic
AI/ML:	AWS Bedrock, Claude, ChatGPT, Retrieval-Augmented Generation